INTEGRATION SOLUTION DOCUMENT

Version 1.0

Confidential

1.Objective

To implement a real-time, direct integration between HRMS (Rymnet) and TMS.
The integration ensures that whenever employee or related data is created or updated in HRMS, the same changes are immediately reflected in TMS through a secure POST API call.

2. Integration Flow

Step 1: Event Trigger in HRMS

HRMS generates an event whenever:
- A new employee is created.
- An existing employee's details are updated.

Step 2: Data Preparation

Relevant employee data is extracted and transformed into the required JSON payload format.

Step 3: API Call to TMS

HRMS sends a POST request to the TMS endpoint with the prepared payload.

Step 4: TMS Data Update

TMS validates the incoming payload, updates its database accordingly, and returns a success or error response.

Step 5: Error Handling

Failed API calls are logged and retry mechanisms are implemented for transient failures.

3. Data Flow

HRMS —> Event Trigger —> Transformation Layer —> Secure POST API —> TMS

4. API Specifications (TMS Endpoint Example)

Endpoint:
https://tms.com/api/hrms/employee

Method:
POST

Headers:
Content-Type: application/json
Authorization: Bearer <Access_Token>

Sample Payload:


```
{
  "staffnumber": "011130203",
  "StaffName": "surya",
  "Address": "123 Main St",
  "Email": "test11@example.com",
  "Phone": "9876543210",
  "IdCard": "ID123456",
  "CurrentPosition": "Developer",
  "DateJoined": "2023-01-01",
  "Status": 1,
```

"Photo":"iVBORw0KGgoAAAANSUhEUgAAABsAAAAqCAYAAABC8iBdAAAAAXNSR0IArs4c6QAAAARnQU1BAACxjwv8YQUAAAAJcEhZcwAAEnQAABJ0Ad5mH3gAAATcSURBVFhH7ZZtaFtVGMf/5yZpbt7atEkb26VrHXbR1Sl0q1Z8W1m1oyqslaH9YEURdSCbgqDClPnJKfTjwH1wIr5ABdkEx1D3YSCMoc7OddRS7MtaY3PbJE3SNkmTptfn3Hvz2sa12PWD9AeH55znnnv+5zn3uc+9TCawSQia3RS2xDaELbEN4f8rtmoF+XNKwuVFGV5mxFKKHMvU+CzeqK/cQdamE9CwKGGfpw5m0UjOfydPrPePCXwkbsNckgZcZEmzWl/O6edauyCjIe7DqSYBd9XVkHN1FLGJUBjtYStuxMlTuCA1uWBxtU97zPRV69QDHbYgTh2oIsdK2KRPkvdGHIgkaJRzc7EokCKRPFFtrubT0fE26X346UU3OfJhjcMpeTSWXSAdRYedoWXiEiwlBnKSL904Wj9icuHzQC1GZmmQswGB2sctfvQ8cBufnYGZrtB55Oxu//I0PrnHDFd5mTrjJgyNedHZF8KY/nZarUQVpHW63H581ZMvJsj0nORF6pF9mDLr24dcaxb6yzeNzm8CGJHCWJ68DkTpWcQpSsrkeuO8NiuLAC5ETaB5p3cboKN0XgteEnr8bACj9LBZiY6iSmJ5ZlDZeI3I8P7T9drMLAKPiO9mnw1wuypV702YlPw4cCmOMX5kIqWgUQdmJEs73m4Grh1JwqCncQEUGQ8buN8Q0VxZBqZ86BoJo/X6DC4Mjyu+MZ+EwzEzFhvtcLRWw3QfpTkXIsHaKhP6X1+C1WJS5i4sRBWbRj1Giq5kYkD1aEjBIO6dSOHMuB8XRyV09Pvx4W8jeC1RhllKb4sgwEJHbqs2Q+cQ4a40Y/DNxjyh5g946FmyCcJLUg5fDAUg+8NAIAT4QxBjcfxa6YaeCTCTCG9czKIXUOcy4+oLdTCWUDYSC9Eo7nwriOnZmDJOk0mQQjGPkXZBIvDPojSRwLPtLbAa9CTCYNai4tZFYpfbKmCzWpT7uJDn6DikwAKQ5AtnyRxjodiTe3aj22WgxXR47uCjMJGQiRbnLR1ZpUGHT+0JlNms2l3Azpd/hzQzT8WBFl0qFONC3MerQgFftu/F8c5W6EmQ0dggMIgkYiJbQbbXNI8djgp1MhGJzGF6Zo5EaFGKSk7xGphFy0ZqBZFxGGN4xSnDo21EESSfnYTeM4Sxw+lQL2jwaUpESR4VNYHevxwEnhw8SVLW7ZorHy74ao6gSO0IC6K+QCgDiahHGAcTSzWnSuYYh8Mrq3QaLnhIH8Ld9Ok7jEBRoShlrJIUXDCVhM7q1K6oqAlCR3umX0aMTy6Cs9yOlypZ8YiIvu+kTGII7j1wWvnBZxHSz2qRKvWJc/lv/HqIRmN4++T39DwSEHa2gVkcONRwQ7uqkld1T1y0ofc8vVvrJBAM4cFnTmPJsg365udJSD2+N56qVmwaVno0Icf5hy+HXfQZ6t71N1pqYxCN9PEswlQoib4fvTg7VIZUhYdSlaePyrG2Obx7MPtacNiFn71yx2drq/Zrpa6CYei4AIEKQC7C/uYafN09pQ3/Ox4Xw9V3kiuEOJlfucGxGRz7oRznBrQXap1QicTJrjh6HrGsKsRZ8ZN6bdiH81ck/DLXCB8Vff4zVQyRHucdpUE84YnhsaYqWMzq56UYK8RuJavHe4vYEtsQNlEM+AdtaP3Ho4axhQAAAABJRU5ErkJggg==",

"PhotoFileName":"AGcruel.png",

"dob": "1995-06-15",

"Extension":"png",

"career_enhance_details": [

{

"career_id": 5551515,

"startdate": "2025-06-09",

"enddate": "Current",

"department": "Security",
"position": "Security Assistant",
"base_data": "PEN",
"division": "6",
"division_text": "Security",
"isactive":1,
"tempEndDate":""
}
],

"education_details": [
{
"id": 22466, // Add actual ID if available
"startdate": "2025-09-30",
"enddate": "2025-10-10",
"institution": "IIT1",
"qualification": "UG1",
"isactive":1
}
],

"experience_details": [
{
"id": 19049, // Add actual ID if available
"startdate": "2025-09-30",
"enddate": "2025-10-11",

"company": "Aventra Group 1",

"position": "Full stack Developer 1",

"isactive":1

}

]

}

- **Photo Format:** base64

Sample Response (TMS):

{
"status": "success",
"message": "Employee record updated in TMS",
"employeeId": "EMP12345"
}

5. Solution Breakdown

Solution 1 – Initial Sync via Excel Upload

Objective: Enable bulk transfer of employee data from HRMS to TMS using Excel.

Process:
1. TMS team provides an Excel template with required columns (e.g., Staff No, Email, Division, Department).
2. HRMS team fills in employee details in the template and returns it.
3. TMS team validates the file and:
   - Updates existing records.
   - Inserts new employee records.

Solution 2 – Initial Sync via GET API

Objective: Automate synchronization using an API-driven approach.

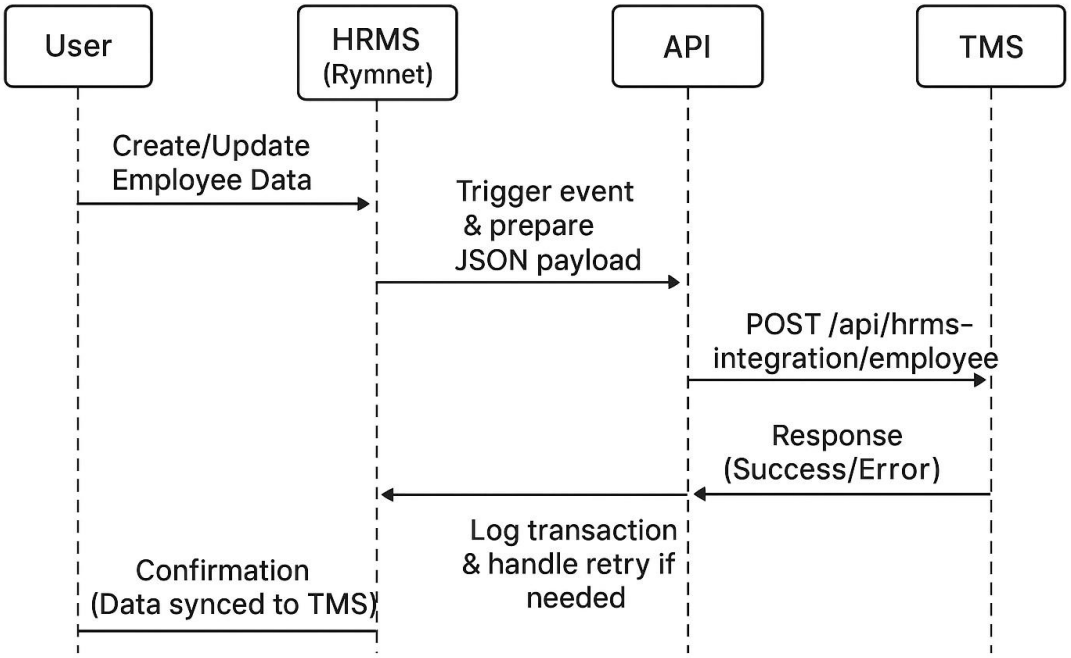
Process:
1. HRMS exposes a GET API that provides employee data.
2. TMS system:
   - Calls the API to retrieve employee data.
   - Compares retrieved data with its own database.
   - Updates existing records or inserts new ones.
3. TMS sends a sync report (success/failure) back to HRMS.

6. Real-Time Update (Post-Sync)

After the initial sync (via Excel or API), the integration switches to real-time mode.

Whenever a new employee is created or existing employee details are updated in HRMS:
- An event is generated.
- A POST API call updates TMS immediately.

Sequence Diagram



1. Data Sync Column

Details from HRMS (Rymnet)	New Recruit	Resignation	Change in Details
Status (automated)	Active	Inactive	–
Staff Name	✓	–	–
Staff ID	✓	–	–
Photo	✓	–	✓
Date of Birth	✓	–	–
Address	✓	–	–
Identification Card (IC)	✓	–	–
E-mail	✓	–	✓
Phone Number	✓	–	✓
Current Position	✓	–	✓
Date Joined	✓	–	–
Educational Details	✓	–	✓
Prior Experience	✓	–	–
Career Experience	✓	–	✓

Legend:

✓ = Field is required to be inserted or updated.

"–" = No action is needed for this field based on event type.